in the Bayesian framework that we had been talking about and started talking about one form of inference in the frequentist framework, and that's maximum likelihood or maximum likelihood estimation. And so we gave this definition here. So we said the maximum likelihood estimator is defined as the argument theta, sort of the input theta, that maximizes the likelihood function, right? So it's not the value of the likelihood function, it's the value of theta that gives us the maximum. And so if you visualize it looks something like this, right on the y-axis, we have the likelihood. On the x-axis, we have the different values of the parameter theta that give us some value for the likelihood. And if we go ahead and find the maximum of that likelihood function and go down to the x-axis at that maximum, right, that value of theta that gives us the maximum, that's our theta hat or our maximum likelihood estimator or estimate if we have realized values for the data. And that's really all there is to it. It's an optimization problem. We just want to optimize that likelihood and find the value that gives us that optimum. We talked about a specific procedure that can be used oftentimes to do this optimization problem, right? Just relying on calculus. And so oftentimes it's convenient to work with the log likelihood rather than the likelihood function. And so this procedure starts by just taking the log of that likelihood function. And then if we want to find an optimum point, We're going to differentiate that log likelihood, set that equal to 0, and that gives us a candidate point. And then we need to check to ensure that the second derivative at that candidate point is negative so that we're getting a minimum or a maximum and not a minimum. But, again, that's not the definition of the maximum likelihood. This is. And from there it was just examples. So we looked at a few examples where we used this procedure to find a maximum. And then we looked at a couple edge case examples where that procedure. doesn't really work for us.

And we commented that this often happens when the support of the data depends on the parameter. So in this example here, the support of the data, the allowable values for X depends on the parameter of theta. It's uniform between zero and theta. So as theta changes, the allowable values of X change as well. So this often leads to these edge cases where that kind of calculus optimization procedure doesn't really work. Maybe the maximum falls on a boundary, right, in this case, the maximum of our observed X values. And similar for this other example, right? If we plotted it, it looked something like this. So this is our likelihood function, and the maximum falls on this kind of boundary point right here. And we get the sharp jump, do the indicator function in the likelihood. Okay, so that's maximum likelihood. We're going to keep talking about maximum likelihood today about some of the statistical properties of this estimator.

So again, I mentioned maximum likelihood is one way to do frequentist inference, if not necessarily the only way. There's other ways like method of moments, so people can calculate the sample mean and the sample variance and match those up to, you know, the mean and the variance of a distribution given by the parameters. Like we know the mean of a beta distribution is alpha over alpha plus beta. Right, so you could set up a system of equations that way. not really going to talk about other methods like that. We're just going to talk about maximum likelihood estimation. But we want to understand some of the properties of this estimator, some of the things that are nice about this estimator, or maybe some things that would be nice that it doesn't have.

What would we like in an estimator? Those are properties. So the first property is what's called invariance.

So this invariance theorem says, if we have a maximum likelihood estimate, theta, half, of some parameter theta.

But maybe we're not interested in theta itself, but we're interested in some function of it, called that G of theta. So we have a maximum likelihood estimate for the parameter theta, but we really care about G of theta, a function or transformation of the parameter. Well, all this invariance theorem tells us is that the maximum likelihood estimate of G of theta is G of theta hat. We're just plugging in the maximum likelihood estimate for theta into that function.

Okay, that's really nice because we only have to do this maximum likelihood procedure one time, get an MLE for theta, get theta hat, right? And then we get the MLE of any function of it, G of theta, for free. You're just plugging it in. There's no extra optimization involved. And we know a lot of different quantities of interest can be functions of the parameter, things like the median of a distribution, might be a function of the parameter, means, might be a function of the parameter, quantiles, probabilities. These are all functions of the distribution parameters. And so we can get the NLEs for those functions for free after we get the MLEs for the parameter. So that's the invariance property of maximum likelihood. And let's look at a couple examples. So let's say we have a random sample of data x1 through xN. And these are wait times for a customer support center. And let's say we model them coming from an exponential distribution with the parameter lambda. But we're asked for the MLE of the average weight time, not necessarily the parameter lambda. Okay, so let's write down what we know. We know that the data, XI, IID, random sample from exponential distribution with some unknown parameter lambda. What's the mean of an exponential distribution? It's one over lambda. Expectation of any X, any random variable, any observation here, is one over lambda. It's the same for all of our X observations because they're IID. So the mean or the expectation is not lambda, but it's a function of lambda. It's one over lambda. So in our case, G of lambda, the transatlantic. of the parameter that we care about is 1 over lambda. So our invariance property of maximum likelihood tells us once we get an MLE for lambda, plug that in to this function, 1 over lambda hat, and that will be our MLE for G of lambda or for the average weight. So that's all we need to do is get an MLE for lambda and then plug it into that function. We know how to get an MLE now. We write down our likelihood function, L of Lambda, given the data X, We have a random sample, so we know it's going to be a product for all of the observations, the data points, of that exponential PDF. Lambda, e to the negative lambda. We do some simplifying, collecting terms. We get lambda to the n times e to the negative lambda, the sum of our X values. And I'm going to work with the log likelihood because it's more convenient. So I take the log of this and the log likelihood function, N log lambda minus lambda, as the sum of the X's. Okay, we have our log likelihood function. We can differentiate it now with respect to our parameter lambda, derivative, this log likelihood function, and that's n over our candidate minus the sum of our observations, our X's, and we're going to set that equal to zero and solve for our candidate lambda hat. And doing so gives us lambda hat equal to n over the sum of the observations. And we would want to take the second derivative to verify that this is indeed a maximum. I'm not going to do that here. we saw plenty of examples of that last Thursday. In fact, I think we did the same exact example. But we found Lambda hat. We found our maximum likelihood estimate of lambda, and it's n, the sample size, over the sum of the observations X. Okay, but that's not what we want. We don't want the MLE of lambda. We want the MLE of the average weight time or one over lambda. So we just plug it into that function. So the MLE, the expected value of X or the average waiting time, which is one over lambda. is one or lambda hat, sum of x divided by n. Question, yeah. This, that's just the word so. Okay, so very nice property, right? We can get the MLE of any transformation of our parameter directly, plugging in the MLE of the parameter itself into the function. Okay, let's do another example. Again, we have a random sample, X1 through XM, again, of these wait times from a customer support center. We're still modeled from an exponential distribution with parameter lambda. And now we want the MLE of the variance of the waiting times. So the problem setup is the same. We're going to have the same likelihood, function, same MLE for lambda. But now the thing we care about estimating is not lambda, and it's not the mean. It's the variance of the waiting times. We still know the data xI, ID exponential. And what's the variance of the exponential distribution? 1 over lambda squared. So the variance is 1 over lambda squared. G of lambda, the function we care about is 1 over lambda squared. We already determined that lambda has. our MLE of Lambda is N for the sum of our observations, the sum of X, right? Now we just plug that in.

So the MLE of the variance of X, which is one over lambda squared, is just one over lambda hat squared.

The sum of the x is divided by n squared, the square of the sample mean.

This, so this is saying G of lambda, our function or transformation of the parameter that we care about, G of lambda is one over lambda squared.

Let's do one more example.

Say we have data x1 through xN. It's a random sample now from the uniform distribution over the interval zero to theta, where theta is a positive unknown parameter.

And we want the MLE, not of theta, but of the median of the distribution.

We found the MLE for theta itself last time, but now you want the MLE for something different, the median of the distribution. Does anybody remember what the MLE for theta was from the last lecture?

Yep. It was the maximum observed X value.

The call Theta hat, our MLE of Theta, is the maximum observed X value, X1 through XN. So we can use that. We're going to use our invariance property here. We just need to know what is the median of the distribution. Does anybody remember how to find the median of a distribution?

The population median, so the median of this entire distribution, yeah. Yeah, you set up a system where you use the CDF at 0.5 or integrate the PDF and find the value that gives you that integral equal to 0.5. So that's what we're going to do. So the median, which I'll call M, is such that integral from 0, M of the probability density function, little F is given theta X, evaluates to one half.

In other words, the CDF evaluates to one half. We want to find that value M. We have one equation, one unknown. We're just solving for M here. Solving for M gives us the median.

So we just need to plug in this probability density function. little f, which is what for this distribution, or theta, yep. So we're integrating from 0 to m of 1 over theta, the x, x, theta evaluated at the limit, m in 0, m for theta. And we're setting that to be 1 half. We want the value m such that that integral gives us 1 half. So we have m over theta equals 1 half, or m equals theta over 2. That's our function g. G of Theta is equal to Theta over 2. We already know the MLE of theta, it's the maximum observed X value, and we have our function G of theta, it's theta over 2, so we can plug in Theta-Hat M-L-E into that function. M-L-E of the median, M-L-E, Theta over 2, a transformation or a function of theta, which is Theta-Hat over 2, or the maximum observed X value over 2. So that's the invariance property, very nice property of the maximum likelihood. estimator tells us, even if we're not interested in estimating the parameter Thea itself, if we're interested in estimating some function or transformation of it, right? We get that for free plugging in our MLE of the parameter into that function itself. Yeah, question? Yeah, instead of solving for MW one-half, you would do one-fourth or three-fourths or any other quantile of interest, yeah. Yeah, exactly. Yeah, the median is defined as the quantile, that's 50% probability mass to the left, 50% to the right. So that's invariance. And the next property I want to talk about is called consistency. So this consistency property, this consistency theorem says, let Theta hat sub-N now, so using a sample size of N, be the maximum likelihood estimate or the MLE of some parameter theta, using a sample size of N. And we're specifying the sample size N here because this is an asymptotic result. So we're going to think about what happens as our sample size gets bigger and bigger and bigger and goes to infinity. That's why we have Theta hat sub-N. It's an asymptotic result. So Theta hat sub n is our MLE with sample size N, and let Theta sub T be the true but unknown parameter value. So this consistency theorem says that as our sample size N tends to infinity, theta hat sub N, our maximum likelihood estimator, converges in probability to the true value. Does anybody remember from stat 131

what it means to converge in probability? Sort of. So the actual definition is that the limit, our sample size, tends to infinity, of a probability. And this probability is based on the distance between the two things. So the distance between Theta hat sub n, our MLE, the sub t. So the probability that this distance is small, smaller than some value epsilon, probability is one for any positive epsilon. So we're looking at the distance here, are these two things close together, is our estimate close to the true value? And if that distance is smaller than some value that you pick, say, epsilon, call that a good estimate, right? And it's saying the probability that you have a good estimate or a close estimate goes to one, any epsilon, any positive epsilon that you like to specify. Epsilon can be as small as you want, but that probability is going to go to one if you have convergence probability. Convergence probability, yeah. You don't ever know it in practice. You can show consistency without knowing it, right? depending on the setup that you have. And so this theorem is telling you, even if you don't know the true value, you do know that your MLE is going to tend to the true value as your sample size grows. Just like the law of large numbers says that you might not know the population mean, but you do know the sample mean is going to tend convergent probability to the true mean, even though you might not know that true. So that's what this is saying. Yeah, question. Yeah, so it can be any positive epsilon. So basically the argument is make that as small as you want and the probability of being close to the true value will always be one for as small as you make that epsilon value.

So the idea here is basically as we get infinite data, we are going to the truth, right? So in some sense, we can kind of trust our estimator. We would never actually have infinite data in practice, but if we did, our estimator would be kind of perfect, right? And so that's, that helps us trust it in some sense. It's some property of the estimator that we can feel good about. It doesn't mean our estimator is going to be the best with a finite sample size, but if we had infinite data, it will go to the truth. Hopefully it seems better than if you had infinite data and you didn't go to the truth. That's the idea with consistency. And so this is just telling us that the MLE is consistent. Related to this is bias. And so bias is maybe thinking about not infinite sample size, but for a finite sample size like you might have in practice. So what do we mean by bias? Well, the bias of an estimator is the difference between its expectation, or expected value, and the true value. So on average, what is our estimator giving us, and how does that compare it to the truth? If on average, it's equal to the truth, there's no bias, but if on average it's equal to something other than the truth, then it would be a biased estimator. And in particular, we define the bias in this way. It's the expectation of the estimator, the long-run average of the estimator, minus the true value. Theta. Consistent if the bias goes towards zero as you're going? The bias does have to go to zero for it to be consistent, but that alone would not make it. You could set up counter examples where the bias goes to zero, but it's still not consistent. But certainly if the bias does not go to zero, it's not going to be consistent, because you will have epsilon, you'll have probability of an error greater than epsilon that's not one. Okay, so, yeah, question. So, yeah, we'll look at this example in a second. and I think that'll help clarify. You wouldn't actually know the true value in practice, but you can think about what happens, you know, under a model, a class of models. In this case, we're going to look at the uniform model. Okay, so if the bias is zero, we call an estimator unbiased. If the bias is anything other than zero, we call it a biased estimator. And it turns out that the MLE, the estimator we're studying here, is not always unbiased. In some cases, it can be, but it's not always unbiased. It kind of depends on the model that you're using. And especially for small sample sizes, bias might be in some sense large. But we did see that the MLE is consistent, right? It is converging to the true parameter value as that sample size tends to infinity as we collect infinite data, right? And so like you just said, this means that that bias has to tend to zero as our sample size grows. Okay, let's look at an example here, a bias example.

Okay, so we have theta x1 through xN iID from the uniform distribution on the interval zero to theta, where theta is our positive unknown. parameter.

We've already calculated the MLE for Theta, right? It's the maximum observed X value. We don't know the true parameter theta here, but we want to know, is our MLE theta hat unbiased? Calculate the expected value of Theta hat.

We need expectation of Theta hat, our MLE. We know theta hat the MLE is the maximum observed X value.

So if we want the expectation of it, we need to understand the distribution of this maximum.

To get this, You need distribution Theta hat or the maximum observed X value that first. So we're going to get this distribution by working with the CDF. We're going to start by finding the CDF, the cumulative distribution function for this maximum. Let's let the random variable Y be equal to the maximum or observed. This random variable Y.

We want the CDF of Y. Capital F question. We know if we differentiate the CDF, we get what?

The PDF. So we know we can get from the CDF to the PDF, and in a moment I'm going to show you why working with the CDF is convenient. It's going to rely on our random sample property or the property of independence here. If you looked at order statistics at all when you took stat 131, you did the same exact thing. You're finding the distribution of an order statistic, right? The maximum is a type of order statistic. So capital F of Y is our CDF, right? And we know from the definition of the CDF, it's just the point. probability that this random variable y is less than or equal to some specific value little y. This random variable y is the maximum x value.

So if the maximum x value is less than or equal to this value little y, then every single one of those x is has to be less than or equal to little y, right?

This one is less than or equal to little y, then every single x is less than or x1 less than are equal to little y, x2 less than equal to little y through xan less than or equal to little y.

The max is true. It's true for all of them. So now we have a joint probability. It's a joint probability statement, but we have independence.

We know x1, x2 through xan are independent. It's a random sample. So if they're independent, what do we know about this joint probability? What's that?

Yeah, it's just the product of the individual probabilities. Independence lets us break it apart like that. So this is equal to, equal to, due to independence. My notation above the equal sign just means due to independence. This is equal to probability x1 is less than equal to little y through the probability xN less than equal to little y. And hopefully you recognize that, again, because it's iID, all of these probabilities are the same. Probability for x1 is the same as the probability for x2 and so on. But what is the probability for one of these? What is the probability that x1 is less than equal to? some value little y.

How would we get that? Yeah, this is the definition of the CDF. But what if we, so far, we only have written down the PDF for this uniform distribution, right? We said it was 1 over theta.

Could we use the PDF to get this probability from 0 to Y? Yeah, we'll integrate the PDF, and that will give us the probability that the random variable X1 is less or equal to the value little Y. And again, it's the same for each of these 1 through N random variables. For each one, we're going to integrate from 0 to y, probability density function, which is 1 over theta. And because it's the same for all end of these, right, we just have this term n times or to the nth power. When we integrate this, we get y over theta. So we have y over theta to the nth power. I'm going to write here for y between 0 and theta because we know past theta the CDF evaluates to 1 and before 0, the CDF evaluates to 0. Yeah.

So we're integrating 1 over theta with respect to the random variable X, right, which gives us X over theta, evaluated at the limits Y and zero, gives us Y over theta. Little Y is just the value the notation we're using for the CDF of the maximum.

It's just an arbitrary realized value, right, in that CDF. Big Y is the max.

Little Y is just some arbitrary value that Big Y could take or could be less than. Only in this expression, it's greater than you're equal to big Y. But it's just like when you write down the CDF, the CDF by definition, is a function, a function of some input, and this is the input to that function. Little Y is the input to that function. So it just tells you for that input, what's the probability that the random variable is less than equal to that input? The input can be whatever we want. We could plug in, you know, one-half, we could plug in three-fours, whatever we want, it can vary. It's just the input to the CDF, though. Yeah, I know the notation can get confusing because we're dealing with N random variables here. We've redefined another random variable as the maximum. We're calling that one Y. And then we're also differentiating between the random variables themselves, the capitals, and then just arbitrary values the lower. So there's a lot of different pieces there. Yeah, question. So we're going to differentiate this in a moment to get the PDF, which is what we actually care about. But notice that because we had the CDF, of Y here, and we had this joint distribution or this joint probability statement, because we, the CDF is a joint probability statement and we have independence, it factors into this product. And that's the nice thing about using the CDF here, is this kind of factorization into this product. Because once we do this, right, this is the same as the next one and the same as the next one, right?

And so now we have CDF for the maximum here. And we got there from that factorization from the independent.

Very often if you have, if you're interested in a PDF of something involving multiple random variables, it can help or it can simplify things to start with the CDF. And if you can get the CDF, then you can differentiate to get the PDF. It's not the only way to get there. You could do like the transformation approach that you just learn in stat 131, but that can get really messy. This was a lot easier to get there. So definitely not the only way, it works out nicely here.

Okay, so we have the CDF now for Y, the maximum observed X value, but we're going to want the PDF because we want to calculate the expected value. We know we need to integrate X times the expected value or Y times the expected value. So we're going to get the PDF next.

PDF, a little F of Y. Right, and we get that just by differentiating. D, Y, of the between zero outside of that interval, the PDF is zero. There's no probability mass. So we started with the CDF for Y or the maximum, and we differentiated to get the PDF for Y or the maximum. We actually care about. Expected value, expectation of our MLE. So we can use this PDF now to get the expectation. Expectation of Y. We know just from the definition of expectation, right? We just integrate the random variable times its probability density. function and we're integrating over the bounds zero to theta here the support that's just the definition of expectation so plugging in we have the integral zero the theta n times y to the nth power now n minus one terms from the pdf and we had one extra y term from the definition expectation or theta to the nth power d y this is n n plus one is one or theta to the nth power is y, n plus 1 power, evaluated at the limits, theta, 0, which gives us n or n plus 1 times theta. Which part? This? Yeah. So this is our PDF, right? This is little F, and then we just have this extra y term, right? So this becomes y to the n instead of y to the n minus 1. So notice everything we've done so far, all of this, this is all stat 131, all the background probability theory stuff. We haven't used statistics at all here. So that's why I really emphasize the importance of reviewing the background material, reviewing Stat 131, because we kind of jump

right in, and we use it a lot throughout the course. The actual statistics side of things usually isn't so bad, right, but we have to rely on the mathematical details from Stat 131 to get there. So everything we've done so far, stat 131, probability theory, right? We cheated a little bit because we already knew the MLE from our last lecture. We already knew the MLE was the maximum. But outside of that, everything here was stat 131. But now we can actually answer the question that's relevant to this course, is Theta Hat unbiased? We calculated its expected value. It's this. So is it unbiased? Unbiased? Consistent. But what does unbiased mean? The difference between the expectation and theta gives us our bias. So it's biased. The expectation of Theta hat is not Theta. For large N, it's very, very clear. close to theta, but it's not theta, right? So we do see the bias tends to zero. As we get more and more data, right, this becomes closer and closer to unbiased, but for any finite sample size, it is biased because this expectation is not theta.

Depends on the context. Yeah, here, you know, your maximum always has to be less than theta, right? So you know it's going to be negative. There's no way to get a maximum of your observed data values that's greater than the true maximum. It just can't happen in this setup. So in this context, it doesn't really mean much. In other contexts, it could. Are you overestimating or underestimating?

How does that change things? So situation-dependent. But here we have n over n plus 1 times theta is our expectation of our MLE, and that's not equal to the true parameter theta.

The theta hat is biased. And I think you had a question earlier about how can we tell if something's biased or not when we don't know the true value of theta, and that's exactly what we did here. We never knew the true value of theta for this problem, but we can get the distribution of the estimator, the distribution of theta hat that depends on that true value, it's a function of that true value, and we can see clearly that it's not equal to the true value, whatever the true value is. It can, yeah. You can have an unbiased estimator, and MLEs can be unbiased. They're just not necessarily always unbiased. So this is an example where you do have a biased MLE, You can and should look and find other examples where you have an unbiased MLE. You can have an estimator that gets better as the sample size grows, but it's still not consistent. Imagine if the bias is 1 over 1 plus N, right? As N gets large, that bias gets smaller and smaller, but it never goes to zero. So there's still no, you would never say less sample size is better, but you still don't have a consistent estimator. Yeah, that's a really good point. You can correct for this. In this case, we found the bias explicitly, so we could just add that bias back and get an unbiased estimator. It's no longer the MLE, but it would be an unbiased estimator. So those are the properties of the MLE that we're going to discuss here. We're going to do another extra credit activity. So put all of your electronics, cell phones, laptops, tablets away, take out a piece of paper and a pencil. You can use any paper class notes that you have. you can use the course textbook. Can you receive like some sort of EDRC? You should get a calendar invite, but they might not have sent them out yet. Okay. Yeah, because it goes through Baskin. Usually I feel like I get it like a few weeks before. Yeah, I feel like Baskin tends to do it more like a week before. I think they try to like triage them almost. When should I reach out to you in case I don't? Next week. Next week? Waits next Tuesday, then. Maybe by the end of this week? Yeah, maybe by the end of it. Okay, so same as last time. You can use your paper notes. You can work together, talk to each other. When you're done, just write your name and the answer that you selected. I'll call your answer and come and put them in this basket. And I'm going to set a 20-minute time limit this time, starting now. You're saying that we're going to do a lot. And then the other, it goes off to get it. So we're saying that we want to turn. You have to find. I'm going to end of the later. I think that it's just X-I. I think that we're seven. Thank you. Yeah. This way. It's kind of a minus x. It's on my desk. It has a deep. Yeah. It's kind of weird action. So, thank for the probably the probably the act. You think you can have a lot. So you wanted the MLU. Oh, sorry, from your perspective. Because it's like this. For x less than theta, yeah. So here's theta, it's less than x. Yeah. So you're saying, is it going like this? We don't

want this. So then the MLA would be like this. Oh, I'm going to be the max. Yeah, but wouldn't it be the max? What would be the max? Not for review of it. I believe it would probably. I think you're not sure. I think it's not appearing for you. When x equal state that is it. When x. When x equal state is where you want to get the model. Yeah. So like a lot of but like, I don't know if we ever, like, I don't know. I don't know if we've ever like the light thing, but, like, like, I don't know. Like, you're saying that we, like, the log, this is, I don't know. It would, but I'm just not saying, like, because we have fed up for like, he hasn't given us like a stamp. Right. It's not the platform. It's not right. I mean, that's for them talking. I don't know. So, I don't see if this is not the right of this right. It's not right. So, I think you're, the same, the only thing is there. Oh, yeah. So we only think the laws for you like to put it in. That would be like if we gave us like X billion through X and then so we need to purchase from the law. The state of what makes in the M. Yeah. So we have to use the data lines. We're probably going to have law. Just to get thoughts. It isn't that. What do my turn? That is the center of the case. So we have actually all right, you know, can you have a little bit more and I just find a lot of it? Yeah, you can actually, wait, you could have to put it data, right, so if we can find out. So, we're general, they generalize that's going to be a far as like L and then. So if we can find, So yeah, so you know, we want to find the light. I think it might be the product of all this number. And then that one there, the data is one by the end. I mean, this one is kind of instead of line. We do. So we define the likelihood. But then we define the likelihood. But then we define the magnitude. I agree. We define the likelihood. What's the way to make it? What's the... What's the... ...and what I think... And I'll be like to... ...you can be a line... ...which is... ...theim not I'm thinking about, I might be like this. ...and... ...and then... ...and then... ...and then... ...the likelihood of the function for this... ...how are we able to know? As we got that, because the general definition is the likelihood. Is this the like we? Yeah, we agree that when you answer? The general definition for likelihood is, we have a budget they have it right? You're not right. This is video. Yes. So if you do like data minus X Okay, ten more minutes. It's a decrease in function because you have B to the other questions. It's not a big to be. It's negative . Yeah, right. The . Is that the . have the . I'm sorry. You can have . Yeah. Yeah. I'm not a sound. Yeah, you're multiplied. Are you making a lot of these? You're not. I did. Wait, when you take the jury, wait, when you take the jury, it is, wait. Wait, so, we can get into N-Data and N-D-N-A, and then that's from the log-up, and the N-Data, and then you can get N-Data, log is, log, you made it to public, you could just the log-in-air. Okay. Well, yeah, essentially the same thing, you just get N-D-Lat-times-Dat-times. is equal to the law. Like a lot, right? And then you can get the river. Do that? Do you get the end? Yeah, and then we'll use the first. So then what we don't know. That means. Yeah. What is the? So we need to find the E. Oh, yeah. So, I'm sorry. And then we get a, what's the E? What is it? What is it? I don't know. I feel like it's just a lot. But we just think that just be the minimum though? Okay, great. Yeah, yeah, yeah. Well, you take another for it's just as possible to see if it's right and then we could determine how to the hand. But we also need to be to be to something like G. So that that makes sense. But maybe it's like now that we use the actual friend. The thing that we just did, Thank you. You're all right. You're all right. I think we're not. I say it's probably signed it and speaking of a week. Yeah, yeah, but this is the maximum like you guys right. It's going to be a comment. Okay. It's going to be so. So then it's going to be that. I don't know. Where it is? Where it is? Yeah. Okay. So, what are you like? I don't know. I don't know. Oh, yeah, I said anything else. I don't know. I don't know. I just went to do. I don't know. I don't know what they would be an option which would be an option guys I would say that I would say don't you can also you have to train this on edge cases Metacognition here, but also also there are more likely. The only strategies are three that are like, it's probably one of those. Yeah. Why didn't... Because there's not... I guess you could say axi, and you think a lot. Yeah, max effects. That's going to go in this direction. Max. Five minutes. Five minutes. Five minutes. I guess we'll be a little with data. The expectation is the means of the final. So we can't have to be right here. Okay. How do you know this plus one? Right. Okay. Yeah, right. I do. Because we have to be it. Oh, we have to be a lot of here. Yeah. This is a lot because, like, who it's like

commercial. Like, you can't have a row? This is going to be the very easy. Okay. I don't know. I don't know. data times, it's just data times and that's just data times that's not you're that should be your having a lot of so we're trying to find the main we're trying to find you in we're trying to find the media We have just like B. Is that exponential? Yeah, the maximum. Which means we're not going to some part. Some of the same thing, they're trying to think, swap that resource to the fight out of the tree right? The minimum value. The minimum is like the maximum of Y. But how would we've written? He doesn't have it. It's great. It's all. Great half of them. We're back to one. We're back to what I think. I'm finding this is a product for what they are. We're going to do that. We have two minutes. We have two minutes right and what's the MLB to be for you? uh, we just reach a lot. Yeah, we are. We have two minutes right to know. If we just took it. You know what? Let's get it. Okay, let's tell us why. Thank you. Like, if we know something about this, I know this time. Have you ever heard of the experiment? Yeah, I'm sorry. It's like it. So we have something we've got to be set up one guy. Let's not say that. We're assuming. We're talking about it. What if it's actually? Not realistically. Probably not. Okay, one minute, make sure to turn them in before I close the lid. Okay, what do you think the two most like the options are? I think it's A or E? A, okay, everyone, I'm going to get the class. Absolutely. I'm going to go over this. Sure. When I went to submit my homework for homework, too, it says I was unenrolled in. Yeah, I think there's an issue with that homework I need to go and fix that. I'll send out a canvas announcement once I do. You're going with this? That's the MLA is the main... Bird mentality! Wait, just like this is one? Yep. What's the answer after? Yeah, I'm going to go over the problem. Okay, anybody else? Last call. Two, one. Okay, so I'm going to go through it. You'll get the answer at the end. We're asked for the MLE of the mean of this distribution. So the first thing you might want to do is find the mean of this distribution. So let's go ahead and start there. We know that the mean or the expected value of a data point X, By definition is just the integral of x times its pdf. That's just the definition of the mean or the expectation. We're going to use that. In this specific case, that is the integral from zero to infinity of x, e to the theta, minus x, d x, e to the theta minus the integral. This should be theta, as the lower bound, theta to infinity, x, e to the negative x, dx. You can use integration by parts to do this. integral here.

So this integral, we all remember integration by parts, the integral A to B of U, DV, U, V, evaluated at the limits minus the integral V.

So in this case, we're going to use integration by parts to evaluate this integral, and it gives us Theta plus 1, E to the negative theta. But that's just the integral here. We need to multiply that by E to the theta to get the full expectation.

Expectation of X is E to the theta. Theta plus 1, E the negative theta is just theta plus 1. So we need an MLE for theta now, and then we know from our invariance property, but that MLE of the mean is the MLE of theta plus 1. So that kind of tells you what the answer is right now. Let's go ahead and complete the exercise. So we need our likelihood function, the thing we like to maximize. The likelihood of theta, given the data x, the product, e, theta minus x, times the indicator, but each of these data values is greater than equal to theta. Theta is a parameter that controls the support of the data, the allowable values for X. And we can simplify this as E to the n theta times E to the negative sum of our observations X. And then we can simplify this product of indicators because we know if every single X has to be greater than equal to theta, that's the same as saying just the minimum of the X's has to be greater than equal to theta. If the minimum is greater than or equal to theta, then they all are. So we can just simplify this as the indicator of the minimum X value is greater than equal to theta. Okay, this is our likelihood function. This is what we would like to maximize. Notice this term doesn't involve theta, constant with respect to theta, so it's not going to impact this function. This term just says theta has to be less than equal to the minimum. So this term is kind of controlling everything, right, E to the N theta. And we see as theta gets bigger, this term gets bigger. bigger.

This is an increasing function. This is an increasing function of theta when the indicator is met.

So if it's an increasing function of theta, we'd like theta to be as big as possible, and the biggest it can be is given by the indicator. The biggest it can be is that minimum. Theta hat, our MLE for the parameter theta, is the minimum of our observed X values.

And then the MLE of the mean, or the MLE, the expectation of a data point X, theta plus one, is Theta hat plus one or the minimum x value.

Now one thing I'd like to point out is, right, this is probably the way I would approach it, but if you're very observant, you might have noticed that this looks a lot like an exponential distribution.

If you let theta be zero, then it is an exponential distribution, the parameter equal to one.

And so by having this theta, we're just shifting the exponential distribution.

So the mean is becoming theta plus one. So you could have skipped the whole integration by parts if you had noticed that. Again, and I don't expect you to notice that. I would have done the integration, but that is a shortcut to this problem. And another reason why it's good to get very familiar with the forms of the distributions, recognize kernels and things like that. It can save time. Any questions here? Yeah, so we have a product of these indicators saying every single X has to be greater than or equal to theta. But if the minimum X is greater than or equal to theta, we know every single other X is also greater than equal to theta. So we can simplify this product of indicators as just an indicator of one thing, that thing being the minimum. It's a way to simplify the product, yeah. So that's it for MLEF for now. It's also everything that will be covered on the exam. So we'll have another lecture on Thursday, but that lecture will not be on the exam. And I will also do some exam review on Thursday. And I'll talk about the exam that'll be next week on Tuesday. For now, you have everything you need to start studying for the exam if you'd like to. Yeah, question. No practice exam, but look very, very good. closely at homework problems and discussion section problems. Those will be very good prep for the exam. Any of the book questions are good practice problems. I don't have specific ones. But yeah, for the sections we've covered, the sections I've listed, any of those problems are good. Correct. It'll be on the next midterm. I think it's like four problems, but they're subparts. So a problem might have like an A, B, C, something like that. Oh yeah, yeah, yeah. 50% is a point. Two-thirds is two points. 75% is three points. All right, see you all Thursday.